

PART A - 10x1=10 Marks - Answer All Questions (MCQ)

- 1 The process of continuously tracking and analyzing the performance and health of cloud resources and applications is called:
a) Deployment b) Replication
c) Monitoring d) Identity and Access Management
- 2 "Identity and Access Management (IAM)" focus on
a) Managing server hardware b) Defining and managing user identities and their access permissions
c) Optimizing network configurations d) Automating software updates
- 3 The key characteristic of a content delivery service is
a) Provides databases for applications
b) Offers on-demand virtual machines
c) Caches and delivers web content from edge locations to reduce latency
d) Manages user permissions and access
- 4 Which of the following is an example of open-source private cloud software?
a) Microsoft Azure
b) Amazon EC2
c) OpenStack
d) VMware
- 5 Which Python statement controls the flow of execution by checking multiple conditions?
a) if-else
b) try-except
c) for loop
d) switch-case
- 6 Which keyword is used to include the contents of one Python module into another?
a) attach
b) include
c) require
d) import
- 7 Which format is best suited for hierarchical and extensible markup of documents?
a) JSON b) XML
c) CSV d) Plain text
- 8 Which Python package is use for image processing tasks
a) NumPy b) boto3
c) Flask d) OpenCV
- 9 Which of the following is a common technique for organizing and segmenting large datasets based on similarities?
a) Classification b) Regression
c) Clustering d) Association Rule Mining

10 In the CSA Cloud Security Architecture, which component is responsible for verifying user identities?

- a) Authorization b) Data Security
- c) Auditing d) Authentication

**PART B- 5x6 =30 Marks - Answer ANY FIVE Questions
(Descriptive)**

11 Explain the concept of replication in cloud computing

12 Compare the three primary cloud service models – IaaS, PaaS, and SaaS.

13 Discuss application services and content delivery services in the cloud. How do these services enhance application performance and user experience?

14 Explain the concepts of objects, inheritance, and polymorphism

15 Illustrate the use of Python's file handling techniques.

16 Explain how Django supports rapid cloud web application development using Python.

17 Explain how Python can be used with Amazon Web Services and explain commonly used libraries

18 Define application performance metrics in the cloud. Explain how these metrics are used to evaluate cloud services.

**PART B- 5x12 =60 Marks - Answer ALL Questions
(Descriptive)**

19 How Map Reduce technique is implemented in Cloud Computing?
[OR]

20 Write in detail about identity and access management services

21 How deployment and Management services are used in cloud computing?
[OR]

22 Briefly explain about Hadoop Job Scheduler and Job execution

23 What are functions in Python? Explain the difference between user-defined functions, and built-in functions with examples.
[OR]

24 Explain the process of file handling in Python. How is file handling in Python utilized in cloud-based applications for data management?

25 Explain how Python codes are used in Web Services
[OR]

26 Describe in detail about MapReduce application using Python?

27 Write in detail about Authentication and Authorization from cloud technology
[OR]

28 How security is implemented in Cloud?